

Sentiment Analysis Report

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1 Methodology

1.1 Tokenizer Selection: BPE vs. WordPiece

For CBOW model, I opted for Byte Pair Encoding (BPE) as the tokenizer. BPE is a data compression technique that iteratively merges the most frequent adjacent pairs of bytes/characters in a corpus. Its primary advantages include:

- **Handling Out-of-Vocabulary (OOV) Words:** BPE can represent rare words or unknown words by decomposing them into known subword units, making it more robust than traditional word-level tokenizers.
- **Vocabulary Size Management:** It balances between character-level and word-level tokenization, leading to a manageable vocabulary size while still capturing semantic meaning through subword units.

WordPiece uses a different merging strategy focused on likelihood rather than raw frequency. BPE was chosen for its straightforward implementation and effective performance in handling diverse text.

1.2 CBOW Model Training

The CBOW model was trained on a truncated portion of the provided text data using the BPE tokenization. The model's objective is to predict a target word based on its surrounding context words. The embeddings generated by this model are then averaged to create document-level embeddings for sentiment classification.

1.3 VADER Baseline

VADER is a lexicon and rule-based sentiment analysis tool specifically attuned to sentiments expressed in social media. It provides sentiment scores (positive, negative, neutral, and compound) based on a dictionary of lexical features. For this analysis, VADER's compound score was used to classify sentiment into positive, negative, or neutral categories.

2 Results and Interpretation

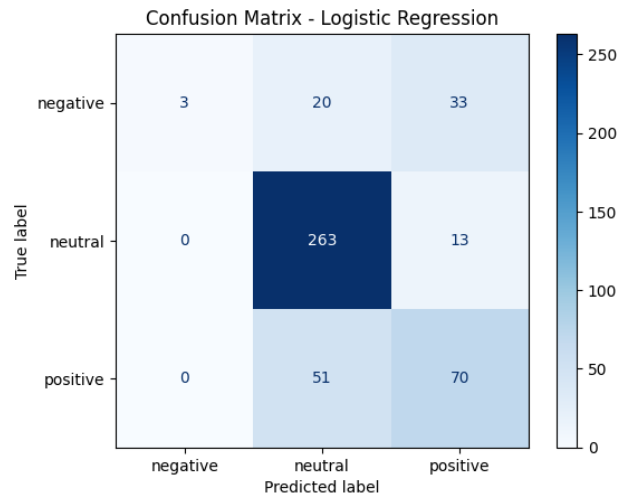


Figure 1: CBOW

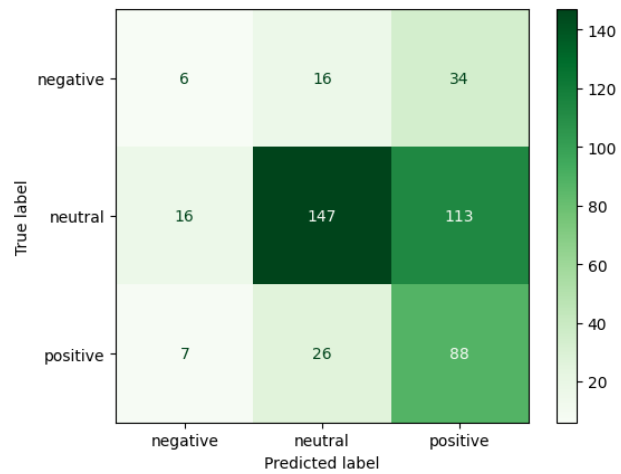


Figure 2: VADER